

A Benchmarking Framework to Disentangle Contamination from True Intratumoural Microbiota in Pancreatic Ductal Adenocarcinoma Identifies Reliable Signals in Fresh-Frozen but Not FFPE Samples

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Abstract

Background: Pancreatic ductal adenocarcinoma (PDAC) harbors a distinct intratumoral microbiome. Yet rigorous characterization of its composition is hampered by pervasive environmental and procedural contamination. Sources of contamination have not been thoroughly explored, and the methods of contamination removal have not been sufficiently evaluated in a benchmarking manner.

Methods: We systematically collected more than 200 negative control (NCT) samples comprising paraffin from formalin-fixed paraffin-embedded samples, lysis buffer, and sterile water, over period of four years processed by different laboratory persons (LP). All samples were sequenced using the whole 16S rRNA gene sequencing with Oxford-Nanopore Technologies. We benchmarked four decontamination methods (Restrictive filtering, Decontam, SCRUb, and the Nejman-derived (Nj) pipeline) against fresh-frozen PDAC samples from tumor-bearing mice, using above NCT survey to calculate a composite score for the assessment. Further, we validated those methods via technical replicas.

Results: Microbial profiles of NCT samples were significantly determined by control type, LP, year, and season, reflecting complex batch effects. The 20 most abundant contaminant taxa spanned well-characterized environmental genera (e.g., *Sphingomonas*) and human commensals from oral cavity (e.g., *Veillonella*, *Streptococcus*). The LP processing samples left a significant microbial trace contributing to the batch effect. Decontamination benchmarking demonstrated that the Nj method consistently outperformed alternatives in both composite score and inter-replicate

concordance. Application of Nj to fresh frozen PDAC samples substantially reduced contaminant burden while preserving putative tumor-associated signals.

Conclusions: Our results support the adoption of the Nj decontamination approach for future intratumoral microbiome studies in PDAC and provide a reliable, batch effect reducing strategy for low-biomass whole 16S rRNA gene sequencing studies.

Introduction

The intratumoral microbiome has emerged as a potentially important component of the tumor microenvironment, influencing immune evasion, drug metabolism, and clinical outcomes across multiple cancer types [citations]. To this end, the microbiome is recently considered as one hallmark of cancer. In pancreatic ductal adenocarcinoma (PDAC), one of the most lethal malignancies, recent studies have reported the presence of distinct intracellular bacterial communities that may contribute to disease pathogenesis and immunosuppression [citations]. However, a fundamental challenge in tumoral microbiome research is distinguishing genuine tumor resident microorganisms from environmental contaminants introduced during sample collection, processing, and sequencing. Low biomass specimens such as tumor samples are particularly vulnerable to contamination, as trace microbial DNA from reagents, laboratory surfaces, and personnel can overwhelm endogenous taxa. Prominent environmental taxa such as *Sphingomonas*, *Ralstonia*, and *Pseudomonas* are routinely detected in negative controls and have been reported as dominant species in tumor microbiome studies that lack rigorous decontamination (4, 5).

Several computational decontamination approaches have been proposed to address this problem, including Decontam (6), SCRuB (2), and the strategy employed by Nejman et al. (7) (Nj). These methods differ in their statistical assumptions, required inputs, and aggressiveness of taxon removal. Nevertheless, a systematic, empirical comparison of their performance in the context of PDAC tumor microbiome profiling has been lacking. Here, we present a benchmarking framework that leverages a longitudinal collection of negative control (NCT) samples acquired across wet lab processing steps, combined with technical replicates of PDAC tissue samples. We evaluate four decontamination strategies using a novel composite score, balancing taxon yield and purity relative to the NCT-derived contaminant profile, and validate findings using inter-replicate Aitchison distances. This work provides practical guidance for microbiome researchers seeking to generate reliable intratumoral microbial profiles from low-biomass PDAC specimens.

Results

1. Contamination Survey in the LaboratoryA longitudinal contaminant survey reveals a lab-specific contamination profile

1.1 Negative Control Samples CollectionNegative controls (NCTs) were collected at every step of the pipeline

To comprehensively characterize the contaminant landscape, we collected NCT samples at each stage of the wet-lab processing pipeline (Fig. 1.A, upper panel), including [78](#) blank paraffin scratches (paraffin controls), [207](#) lysis buffer controls (buffer), and [52](#) sterile water samples used for diluting PCR-reactions (PCR controls), yielding an initial pool of 6,[017](#) taxa. ~~Taxa with extremely low prevalence or abundance were removed (See Methods section).~~ To reduce noise, we applied the following filtering steps: (i) samples with fewer than 500 reads were removed from further

analyses; and (ii) extremely rare taxa were excluded using the following criteria: a detection threshold of at least 2 reads per sample, presence in at least one sample, and a total abundance of at least 1×10^{-6} across all samples, corresponding to 189 reads in our dataset. ~~—and~~ The filtered samples were then normalized via Wrench (8) or rarefaction (9). After filtering, 26203 NCT samples remained: 16713 buffer controls, 7584 paraffin controls, and 206 PCR controls, encompassing 2,2231,775 taxa after rarefaction and 3,073282,305 taxa prior to rarefaction. These controls were accumulated between 2021 and 20254, with a sharp expansion of paraffin and buffer control types introduced in 2022 (Table 1). Relevant batch covariates include processing laboratory personnel (LP), year, and season (Figure 1.A, lower panel1, lower panel).

1.2 Different negative control types harbor distinct contaminant profiles

According to alpha analyses, there is no significant differences regarding the number of observed species among different control types. Notably, Shannon index in buffer control type is significantly greater than one in paraffin controls (p-value < 1e-3), suggesting more evenness distributed of taxa in the first type (Figure 1.B). Nevertheless, when the statistical test accounting for LP, there is no more significantly different (p-value = 0.25), suggesting that LP batch effect may contribute a major role in the contaminant profile. Furthermore, beta diversity reveals that NCT samples are heavily clustered with respect to control types, even after accounting for LP batch effect (Figure 1.C and Supp-sheettable T1 in Supplementary).

~~Intra-sample and inter-sample differences in microbial diversity~~ Microbial alpha and beta diversities were observed across control types (Figures. 1.2 and 1.3). Buffer controls

~~exhibited significantly higher species richness and evenness compared to paraffin and PCR controls, as measured by observed species count, Shannon index, and inverse Simpson index. Analogous patterns were observed when stratifying by LP, year, and season (Supplementary Figures S1.xxx, S1.xxx), suggesting pervasive batch structure in the contaminant pool.~~

NOTES:

- ~~• Year: #observed species (sig. different), shannon index (n.s), beta diversity (significant)~~
- Technicians, seasons: alpha diversity (mixed), beta diversity (significant)

1.3 Microbial Composition of Negative Control SamplesMany environmental taxa are found in NCTs, at varying abundance

Next, to explore the biological origin of these bacteria, we focused on dominant genera in our NCT sample set ~~(figure 1D)~~ and revealed a mixture of well characterized environmental microbes and small portion of human commensals (Figure 1.D). For example, *Sphingomonas*, a ubiquitous environmental genus frequently detected in hospital and laboratory environments, was among the most abundant taxa in nearly all negative controls. In contrast, genera considered as normal flora from the oral cavity, like *Haemophilus*, *Streptococcus*, and *Veillonella*, were ~~both~~ high prevalent ~~and high~~less abundant in the NCTs suggesting an introduction of contamination by the LPs. This coexistence of environmental and human commensal taxa in negative controls requires

a need for nuanced decontamination strategies, since an indiscriminate removal of all NCT-associated taxa risks eliminating clinically relevant tumor microbiome signals.

Compared with the PCR control, the buffer control showed enrichment of several taxa, including *Mesorhizobium terrae* and multiple species in *Burkholderia* and *Ralstonia* genus, while a smaller set of taxa, including *Prevotella histicola* and *Diaphorobacter aerolatus*, were depleted (Figure 1.E). In the paraffin control, enrichment was dominated by species in *Sphingomonas* and *Brevundimonas*, whereas *Diaphorobacter* taxa, *Prevotella histicola*, and *Diaphorobacter aerolatus* were reduced relative to PCR. Overall, both control types revealed a clear mix of enriched and depleted features, but the buffer and paraffin control samples showed distinct microbial signatures compared with the PCR baseline.

~~The abundance and prevalence of NCT in higher taxonomic rank could be found in the supplementary **Figures S1.7-10**.~~

~~Following the correction for batch effects, differential abundance analysis (DAA) among control types revealed several taxa specifically enriched in paraffin controls (taxa **XXX** from the analyses, Figure 1.6), pointing to contamination sources specific to paraffin embedding procedures.~~

~~Need elaborations~~

Table 1. Negative control samples collected across years, stratified by control types.

Sample ↓ /					
Feature →	2021	2022	2023	2024	2025
Buffer	<u>26</u>	<u>67</u>	<u>1724</u>	<u>380</u>	<u>19</u>
Paraffin	<u>25</u>	<u>567</u>	<u>911</u>	<u>12</u>	<u>7</u>
PCR	0	<u>43</u>	<u>43</u>	<u>80</u>	<u>4</u>

2. Technician Influences Microbiome More Than Processing Environment

As we discovered highly prevalent and abundant oral microbiota in our NCT samples, we further explored the possible batch effect that is introduced by the LP that processed the samples. To this end, two LPs performed DNA extraction, library preparation and sequencing of four sterile water samples, which were equally distributed. The experiment was repeated twice in different environments. First, one LP processed samples in a low-germ environment and the other one under standard condition. For the repeated sample LPs switched processing environment (Figure 2.A1; see Materials and Methods for full protocol). Notably, in univariate analyses alpha diversity analysis revealed no significant difference with respect to environmental condition (Figure 2.B, Observed species: $OR=$, $p = 0.694$, Shannon index: $OR=$, $p = 0.733$), but a clear distinction between LPs was observed (Observed species: $OR=$, $p = 0.001$, Shannon: $OR=$, $p < 0.001$, Figure 2.B2). In line with this results, beta diversity revealed significant clustering by LPs identity ($p\text{-value} < 0.001$), with no significant effect of processing environments ($p\text{-value} = 0.3926$; Fig. 2.C3, $p\text{-value} = 0.645$, not displayed in the figure). Notably, after accounting for the effect of LP by including it as a

blocking factor in the PERMANOVA, processing environments exhibited a clearer effect; however, this effect remained statistically non-significant (p-value =0.09). Overall, this indicates that while LP-derived contamination is dominant, environmental setting contributes an independent, detectable signal.

Oral and skin commensal flora mainly drive the LP effect

Next, we investigated the main factors driving the distinct impact of LP on the samples' contaminant profiles. We applied the MaAsLin2 R package with default parameters, specifying processing environment and LP as fixed effects. Notably, the order of these fixed effects is important, as it ensures that differential abundance analyses for LP are performed after adjusting for the effect of the processing environment. These analyses identified a set of taxa consistently enriched under bench conditions, consistent with human skin and oral commensals introduced during opening bench handling (Figure 2.4D, a full list of significant differential abundant species is available in the Supplementary).

3. Benchmarking of Decontamination MethodsA multi-layer statistical testing approach reliably removes contaminants

As previous hallmark publication pointed out, the intratumoral microbiome might play a crucial role in progression of PDAC and the development of chemotherapy resistance. In light of the low-biomass environment, the assessment of intratumoral microbes is susceptible to the presence of ubiquitous contaminants. To this end, a thorough

decontamination approach is necessary to disentangle contaminants from true signals. To identify the most robust method, we performed benchmarking experiments applying four decontamination strategies, including restrictive filtering, Decontam R package (6), SCRuB (2), and multi-layer statistical testing approach such as Nj procedure (7) (Nj for short) to the 16S profiles of eighteen fresh frozen (FF) PDAC samples, generating four decontaminated abundance tables alongside the non-decontaminated data (ND). Notably, the decontamination benchmarking utilizing presumed ground truth has been reported elsewhere [SCRuB paper]. Nevertheless, in extreme low-biomass setting such as PDAC intratumoral microbiome samples, the assumption of a predetermined bacterial composition may not hold. Thus, there is a need to develop benchmarking approaches that does not rely on the ground truth. ~~The comparison of different decontamination methods were done with two different normalization techniques: rarefaction and wrench normalization.~~

3.1 Assessment Using Longitudinal Negative Control Survey

In the first approach, we leverage the longitudinal contamination survey described above as a lab-specific contamination background (CB) for the assessment. Specifically, a bacterial profile, before or after a certain decontamination, along with CB will be used to calculate a corresponding composite score which simultaneously rewards the removal of contaminant taxa in CB while prevents the excessive removal of the true taxa (see Methods and Supplementary for technical details). Scores were computed per sample, and methods were compared using a paired Wilcox test.

~~Firstly, a comparison was made between decontamination data sets and the longitudinally derived contaminant profile. Secondly, we performed additional~~

sequencing on the same samples more than two years later from a different LP to control for the batch effect and investigate which decontamination approach most robustly identifies the intratumoral microbial composition.

3.1 Assessment Using Longitudinal Negative Control Survey

We leveraged the longitudinally derived contaminant profile from above mentioned experiments as a negative reference for evaluation and defined a composite score comprising two complementary components: yield and purity. First, yield is the proportion of putative significant taxa absent from the NCT survey over the number of observed species. On the other hand, purity is the proportion of NCT-absent taxa among all putative significant taxa over the total number of putative significant taxa. The composite score, which is the product of yield and purity, simultaneously rewards methods that exclude contaminant taxa while prevents excessive removal of true taxa (see Materials and Methods section for more details). Scores were computed per sample, and methods were compared using a linear mixed model (LMM) to account for sample level random effects. The residual normality assumption of LMM was validated through quantile-quantile plot (Supplement Figure 3.A). The Nj method achieved the highest composite score, significantly outperforming all alternatives (Figure 3BA, supplemental table ...). ND data scored lower than Nj and restrictive method, on the same level with Decontam method. Interestingly, the composite score from SCRuB is the smallest. Normally, SCRuB shows its effectiveness from the leakage well information, which was not applicable in our experimental settings.

These results were confirmed with datasets normalized with rarefaction (Supplemental figure ..., supplemental table).

3.2 Assessment Using Technical Replicates

To provide an orthogonal validation ~~controlling for the batch effect~~, we re-sequenced ten FF PDAC samples from KPC mice more than two years after initial sequencing from a different LP__as technical replicates. Our underlying assumption was that decontaminated profiles, should exhibit greater concordance between replicates from the same tumor sample than the technical replicates without decontamination (ND).

Inter-replicate dissimilarity was quantified using Aitchison distance, with pairwise comparisons assessed by paired Wilcoxon signed-rank test. The Nj method obtained the smallest inter-replicate distances, indicating superior within sample consistency (figure 3CB, supplemental table ~~tab T4...~~). Consistently, inter-replicate among ND samples showed the highest scores, indicating suspected batch effect.

We also validated the results with other distance metric such as unweighted Jaccard ~~and various normalization methods such as rarefaction and Wrench normalization~~, result in similar pattern. (See supplemental figures S3.C2xxx)

3.3 Test of Ratio

Furthermore, compositional overlap between replicas, measured as the fraction of shared taxa out of total detected taxa between two replicates, was also highest for Nj method (~~Table 2 and~~ Figure 3DE). This measurement estimates the robustness of decontamination procedures. Notably, we excluded the extremely rare taxa by applying a prevalence threshold of at least 2/10 and total abundance cutoff of 1e-4, and we aggregate data into genus rank level. The SCRuB method showed highly variable results, with a few samples exhibiting exceptionally high overlap ratios while the rest were very low. In contrast, the Nj method showed the smallest variation among methods

and produced significantly higher overlap ratios than all others. The full pairwise comparisons are available in the supplementary (tab T5). Other results for rare taxa filtering and different taxonomic rank could be found in the Supplementary (Figures S3.xxx and tables xxx).

Table 2. Overlap: fraction of shared taxa over total taxa among replicates, counted in genus rank, with minimal prevalence of 5% and minimal_total_abundance 0.001%

Methods	Observed	Observed	Count of		fraction
	genera	genera replicate	intersection	union	
Methods	replicate 1	2	intersection	union	fraction
Raw	140	106	71	175	0.406
Restrictive	116	95	66	145	0.455
Decontam	124	102	70	156	0.449
SCRuB	136	105	74	167	0.443
Nj	63	65	50	78	0.641

3.3. Biological plausibility of identified true taxa

Please expand and provide tables.

3.4 Technical replicas with decontamination method reveals reliable microbiome profile for low-biomass samples

We applied above four decontamination strategies to each replicate, along with ND as a base-line to reveal the microbial composition of PDAC intratumoral samples. Furthermore, we took the overlap across replicates as a conservative

estimate of the corresponding bacterial composition. The ND replicates showed a clear sign of contaminants, with the *Sphingomonas* and *Caldibacillus* accounting for a substantial proportion of reads in the first and second replicates respectively. The decontam method failed to remove *Caldibacillus* genus in one replicate; however, taking the overlap substantially reduced the abundance of *Caldibacillus* species (Figure 3.E). Interestingly, a simple taxon-removal strategy based on NCT samples, such as the restrictive method, failed to remove contaminants such as *Cloacibacterium*. By contrast, Nj and SCRuB methods yielded similar compositional profiles. After Nj decontamination, the majority of bacteria belonged to *Ligilactobacillus*, *Escherichia*, and *Alistipes*, which are common gut taxa and may migrate directly into tumor microenvironment, as described in Schorr et al. 2023 (<https://doi.org/10.1038/s41522-023-00446-9>).

4. Intratumoral Bacterial Profile of PDAC Fresh Frozen Samples

We applied all decontamination strategies to characterize the intratumoral bacterial community of the ten replicated PDAC fresh-frozen samples. In the raw, undecontaminated profile, samples were dominated by environmental contaminants. For example, *Sphingomonas* predominated in first replicates of several samples, while *Caldibacillus* was prominent in second replicates of others (see Figure 4.1). These taxa were largely eliminated following decontamination.

Despite rigorous decontamination, the bacterial profiles of technical replicates from the same tumor were not identical, consistent with the known spatial heterogeneity of intratumoral microbiome distributions (14). Nevertheless, decontamination could be able to significantly reduce inter-replicate Aitchison distances relative to raw data (Figure 3.2), supporting the conclusion that residual compositional differences reflect biological variation rather than technical noise. Following Nj decontamination, several taxa with prior biological relevance to PDAC were consistently detected across replicates,

including *Alistipes* and *Bacteroides* which are common gut comensal and their proliferation fluctuated across pancreatic patients [10].

Discussion

In this study, we present a systematic benchmarking framework for assessing decontamination methods in low biomass 16S sequencing of PDAC tumor microbiome. By combining a longitudinal negative control survey with technical replicate sequencing, we establish two complementary: data driven evaluation metrics, a composite yield-purity score, and Aitchison inter-replicate distance which do not require a priori knowledge of true contaminants.

Our negative sample survey revealed that the laboratory contaminant landscape is shaped by multiple interacting batch factors, including control types, processing technicians, year, and season. Crucially, human commensals such as *Veillonella* were present in a majority of NCT samples, challenging the common practice of using any NCT-detected taxon as a definitive contaminant label. Instead, thorough and multi-level filtering such as (7) shows robust ability to selectively remove likely contaminants while preserving taxon diversity, a balance that is critical when studying the sparse, low abundance microbial communities characteristic of pancreatic tumor tissue. Notably, due to the limited number of negative control samples, binomial statistical tests may lack the sensitivity required to effectively filter out false positive contaminant taxa. For instance, a taxon appearing in only a single tumor sample, but absent from the NCTs, would pass the binomial test and be misclassified as a true intratumoral taxon. This highlights a potential limitation in contamination detection when NCT sample sizes are insufficient to capture rare contaminants. Thus, instead the Fisher's exact test was used in our study.

The Nj decontamination method consistently outperformed Restrictive filtering, Decontam, and SCRUB across all evaluation metrics. This advantage likely stems from Nj's incorporation of both prevalence-based and rigorous statistical test, as well as its conservative approach to taxon removal. Our findings align with the original validation of this strategy in Nejman et al. (2020), while extending it to a standardized benchmarking context with independent replication.

Several limitations should be considered. First, the NCT profiles used as reference contaminants may not capture all sources of contamination present in intratumoral samples, particularly those introduced at tissue collection. Second, the small sample size ($n = 18$, and $n=10$ with replicates) limits statistical power for detecting rare intratumoral taxa and constrains generalizability. Third, FFPE samples, which comprise the majority of archival PDAC biobanks, were not included in the study due to restoration complexity and overwhelmed by environmental contaminants, which beyond the capacity of intratumoral microbiome restoration.

Based on our findings, we recommend the following quality control pipeline for 16S sequencing regarding the intratumoral microbiome studies in PDAC: (1) include

negative controls representing all wet-lab processing steps; (2) acquire technical replicates, at least for a subset of samples; (3) consider batch correction along with decontamination; and (4) apply various decontamination method, followed by validation of residual taxa using the composite yield-purity framework and iter-replicates distances described here. Fresh frozen samples should be prioritized over FFPE where feasible, as FFPE restoration introduces additional sources of DNA degradation and chemical modification that complicate microbial profiling.

Materials and Methods

Sample Collection and Processing

Negative control samples were collected at several steps of a pipeline to counter against contamination. Both NCT and tumor samples were sequenced via whole 16S rRNA gene. Dorado (version 0.9) was used as a base-caller, and MetaPONT (15) was used to obtain bacterial profile of a sample. MetaPONT's library comprises complete genome of 13550 bacterial taxa, latest updated on December 2024.

Mice experiments were conducted in the Central [Animal Experimental](#) Facility at the University Medical Center Goettingen, Germany. Approval was obtained by the LAVES Niedersachsen, Germany (application number 19/3085). [Animals](#) were housed under the following conditions: light-dark-cycle of 12 h/12 h, 23 ± 1 °C room temperature (RT), 40–60% humidity, and non-sterile cages. The KPC (*LSL-Kras*^{G12D/+};*LSL-Trp53*^{R172H/+};*Pdx-1-Cre*) mouse model used in this study has been previously described. After tumor onset animals were screened 2-3 times a week and sacrificed when following criteria were fulfilled: overall morbidity, lethargy, signs of pain, rough fur, loss of self-care and social behavior, [ascites](#), [cachexia](#), [body weight](#) loss of $\geq 20\%$.

DNA-Extraction

16S rRNA Sequencing

Normalization

We applied rarefaction and Wrench method in R package (8) for normalization.

Trivial Filtering Taxa by Prevalence and Abundance

In the section of contaminant survey study as well as contamination investigation in various environments, we utilized `taxa_filter` function from R package `microViz` (13) to eliminate extremely rare and possibly noise taxa. The pseudo-code is provided in the technical supplementary file.

In the murine PDAC fresh frozen study which comprises PDAC and corresponding negative control samples, we applied slightly different threshold for each category. For the PDAC true sample, we set the threshold for minimal total abundance as $1e-6$, while for their respectively negative control we reduce this threshold to $5e-7$, due to the assumption that the microbiome content in negative control is much sparser than in tumor sample. Detail and pseudo-code could be found in the technical supplementary file.

Statistical Analysis

In the contamination survey and negative sequencing in different environments, we applied t-test and PERNOVA test along with alpha and beta analysis respectively. Multiple testing correction is Benjamini-Hochberg procedure.

In the murine PDAC samples study, we applied either paired Wilcoxon signed rank test or linear mixed model with random effect to evaluate various decontamination methods. Multiple testing correction is Benjamini-Hochberg.

Decontamination Methods

Four decontamination approaches were evaluated: (1) Restrictive filtering: taxa detected in any NCT samples were excluded from further analyses; (2) Decontam (6): probability-based classification of contaminant taxa using negative control prevalence; (3) SCRuB (2): source tracking-based contaminant removal; and (4) Nejman et al. procedure (7): the decontamination strategy incorporating multiple filtering criteria. Notably, instead of binomial testing as Nejman et al. described in (7), we applied Fisher test instead due to small number of samples and negative controls. The implementation of those four decontamination methods is available in the github repository.

Composite Score

Composite score is calculated from the bacterial profile after or before decontamination and the longitudinal contaminant survey. This score comprised of two parts: yield and purity. Yield is an entity quantifying the percentage of number of putative true taxa not overlap with NCT survey over number of total observed taxa, while the purity calculates the fraction between the same nominator over the number of putative true taxa from a certain decontamination method. The technical detail of the calculation could be found at Supplementary.

Data and Code Availability

Abbreviations

DAA – Differential abundance analysis

| FF – Fresh frozen samples

| CB – Contamination background

FFPE – Formalin-fixed paraffin-embedded samples

LP – Laboratory person

NCT – Negative controls

ND – Non-decontaminated

Nj – Nejman approach

ONT – Oxford Nanopore Technologies

PDAC – Pancreatic ductal adenocarcinoma

SCRuB

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